

Video Analysis Report

I Built a Mesh Network Across the World

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2026-04-12 09:24:56

Video Analysis Report

Field	Value
Title	I Built a Mesh Network Across the World
Channel	Data Slayer
Upload Date	20260412
Video ID	6ibr7gm8uVA
URL	https://www.youtube.com/watch?v=6ibr7gm8uVA
Model	claude-sonnet-4-5
Processed	2026-04-12 09:24:56
Patterns	2 succeeded / 2 total

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Executive Summary

Creating new session: 20260412_092357.process_video.6ibr7gm8uVA
Reticulum Network Range Testing: Building an ISP-Free Internet

Overview

This video documents an extensive real-world testing series of Reticulum, an open-source internet protocol that enables network communi-

cation without traditional ISPs. The creator progressively tests the protocol's range capabilities—from across a house to across continents—using various radio technologies including WiFi HaLow and LoRa, ultimately demonstrating that decentralized mesh networking could provide a viable alternative to centralized internet infrastructure.

Initial Setup and House Test

[00:00:00] The Challenge - Goal: Send messages using only the Reticulum network, no traditional ISP dependency - Test progression: House → Neighborhood → City → International - Core principle: Building a parallel economy of open-source, decentralized, affordable tech

[00:03:15] Hardware Configuration - Base node: Green Haven node (Raspberry Pi running OpenMana fork of OpenWrt) - Features sub-gigahertz 900MHz HaLow chip for extended range - Bridges WiFi, Ethernet, and HaLow interfaces - **Client setup:** MeshChat application running RNS (Reticulum) service - Uses auto-interface discovery for network connections

[00:06:30] Mobile Node Setup - Helltech HD-T01 board running OpenWrt with HaLow radio - Android device running Sideband app (Reticulum client) - Alternative: iPhone with USB-C to Ethernet adapter connected directly to router - Power consumption: Less than 1 watt at 5 volts

[00:11:45] First Messages Sent - Successfully transmitted text messages across the house - **Voice messages** and **high-resolution images** sent over Reticulum - All communication flowing through HaLow radio mesh network

Neighborhood Test

[00:14:20] Antenna Upgrade - Mounted 9 dBi gain fiberglass outdoor antenna outside cement brick house - Used tension mount system to avoid drilling holes - BNC to SMA male adapter for Haven node connection - Antenna positioned approximately 20-30 feet off ground

[00:18:45] Mobile Setup Refinement - Magnetized antenna board mounted on Tesla vehicle - Switched to standard stub antenna for linear polarization - Powered via 12V car outlet with USB-C connection - Android phone connects to Helltech node via 2.4GHz WiFi (no Ethernet required)

[00:21:30] Range Results - 1,000 feet: Messages delivered successfully - **1,400-1,500 feet:** Test three delivered - **2,000 feet:** Signal limit reached with current one megahertz HaLow channel config-

uration - Cement brick houses created significant obstacles for signal penetration

City Test Attempts

[00:23:00] The Challenge Escalates - Target: Send messages to neighboring city (approximately 4 miles) - Initial attempts using WiFi HaLow with kite-mounted aerial node - **Issue:** Intermittent connectivity—node would spot base station briefly then lose connection

[00:25:15] Pivot to LoRa Technology - HaLow limitations in residential setting without high altitude advantage - **LoRa advantages:** - Operates below noise floor - Better obstruction penetration - Adjustable parameters (coding rate, spreading factor, bandwidth) - Trade-off: Lower bandwidth but superior range characteristics

[00:28:40] LoRa Configuration - Two Helltech v4 boards running 27 dBm transmit power - **Parameters optimized for range:** - Bandwidth: 125 kHz - Spreading factor: 12 (maximum) - Coding rate: 8 - Ping script sends automatic connection test every 60 seconds

[00:30:00] Straightaway Test Results - **0.5 miles:** Ping 26 received (RSSI: -44, SNR: 5 dB) - **1 mile:** Ping 27 received (RSSI: -85, SNR: -3.2 dB) - **1.5 miles:** RSSI -92, SNR 3.2 dB - **2 miles:** Ping 68 (RSSI: -91, SNR: 4 dB) - **3.4 miles:** Signal maintained (SNR: -6.2 dB) - **Maximum range:** 3.5 miles with line-of-sight on flat road

[00:33:20] Key Findings - Line-of-sight proved more critical than elevation for LoRa - Metal structures act as Faraday cages, blocking signals - Urban reflection less effective than anticipated - Node wasn't elevated—just on tripod at ground level

International Test: Florida to Venezuela

[00:35:00] The Math Problem - Coast-to-coast distance: 2,500 miles - At 3.5 miles per hop: **800 LoRa nodes required** - Alternative satellite solutions (MeshSat, Iridium) require infrastructure agreements

[00:36:30] The Solution: Mesh VPN Tunneling - **Key insight:** Reticulum is protocol-agnostic—works over any data link - Implementation using **Tailscale** mesh VPN - Creates private encrypted network between distant nodes - Packets travel over existing infrastructure but in encrypted private tunnel

[00:38:15] Existing Workflow Example - Creator already uses Tailscale with TrueNAS for video collaboration - Files dropped on NAS

in Florida appear instantly in Venezuela - Same concept applied to Reticulum mesh networking

[00:40:00] Network Architecture - Origin: LoRa Helltech node in Florida - **Client:** Android running Sideband - **Bridge:** Computer with R-node connected to Tailscale TCP/IP interface - **Destination:** Venezuela computer listening on Tailscale IP address - **Result:** Message successfully delivered across continents

[00:42:30] The Successful Test - Message sent from Android phone via LoRa - Traveled through R-node to computer - Crossed encrypted Tailscale mesh VPN - Appeared on MeshChat in Venezuela - **Entirely over Reticulum protocol**

Technical Advantages and Philosophy

[00:44:00] Why This Matters - Reticulum as “blockchain of communication”: Decentralized, community-run, unstoppable - No single entity owns or controls the network - End-to-end encryption by default (unlike ham radio) - Operates in unlicensed ISM band (unlike ham radio requiring licenses) - Linear scaling instead of exponential overhead (unlike traditional mesh protocols)

[00:46:15] Comparison to Ham Radio - Ham radio requires licensing for power above 4W EIRP - Cannot encrypt ham radio transmissions - Analog vs. digital full internet capability - Reticulum provides privacy, encryption, and unlicensed operation

[00:47:30] The Bigger Picture - Power returning to individuals vs. centralized control - Following American founding principles: government serves people - Decentralization as check against state coercion - Technology democratization through open-source tools

Future Implications

[00:49:00] The Vision - Currently borrowing internet infrastructure with private encrypted tunnels - **Long-term goal:** Replace tunnels with radios, satellites, community nodes - Internet becomes optional, not required - Free decentralized mesh immune from state control

[00:50:15] Barrier to Entry Collapsing - Radios becoming cheaper and more capable - AI accelerating open-source development - Projects emerging that legacy systems cannot approve - “Cyberpunk renaissance” driven by accessible technology

Conclusion

[00:51:00] Mission Accomplished - Successfully demonstrated Reticulum communication across: - ☐ House (short range) - ☐ Neighborhood (~2,000 feet) - ☐ City (3.5 miles with LoRa) - ☐ International (Florida to Venezuela via mesh VPN)

Key Takeaway: Reticulum proves that building an ISP-independent internet is not only possible but increasingly practical. While current tests utilized internet infrastructure for long-haul connections, the protocol's flexibility means those links can eventually be replaced with community-owned radios, satellites, and mesh nodes. The technology is ready—what's needed now is deployment and adoption.

The creator hints at a “bold idea” for future development, suggesting more ambitious projects to come in the decentralized communication space.

Summaries

Structured Summary

Creating new session: 20260412_092357.process_video.6ibr7gm8uVA
ONE SENTENCE SUMMARY:

Testing Reticulum, an open-source mesh network protocol, to send encrypted messages across distances from house-level to international using various radio technologies.

MAIN POINTS:

1. Reticulum is an open-source internet protocol enabling networks without ISPs through automatic mesh networking capabilities.
2. House test successfully sent text, voice, and images across Wi-Fi HaLow using Raspberry Pi nodes and MeshChat.
3. Neighborhood test achieved approximately 2,000 feet range using sub-gigahertz antennas before signal degradation occurred.
4. City test struggled initially with Wi-Fi HaLow due to concrete obstacles and insufficient antenna elevation.
5. Switching to LoRa radio achieved 3.5 miles line-of-sight communication with proper spreading factor configuration.
6. Reticulum is protocol-agnostic, working across Wi-Fi, Ethernet, LoRa, and other data transmission methods seamlessly.
7. Messages are end-to-end encrypted by default, unlike ham radio which cannot legally use encryption.

8. Tailscale mesh VPN enabled international message transmission from Florida to Venezuela over private encrypted tunnels.
9. The technology enables decentralized communication networks independent of traditional ISPs and government control mechanisms.
10. Eight hundred nodes would theoretically be needed for coast-to-coast LoRa coverage without satellite assistance.

TAKEAWAYS:

1. Reticulum successfully enables encrypted mesh networking across various distances using affordable, open-source hardware and software.
 2. Line-of-sight matters significantly more than elevation for LoRa range, with concrete buildings severely limiting propagation.
 3. Mesh VPNs like Tailscale can bridge distant networks temporarily while physical infrastructure develops independently.
 4. Decentralized communication networks offer censorship resistance and privacy that traditional ISP-dependent internet cannot guarantee.
 5. Combining different radio technologies and protocols creates flexible, resilient networks adaptable to various use cases.
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